WELCOME TO

ISLAMIC UNIVERSITY OF TECHNOLOGY

Hum 4147

Technology, Environment and Society

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Technology, Environment & Society

Course Learning Outcomes (COs)

After completing the course student will be able to

CO1: Explain the interdependency among technology, environment, society, and sustainable development;

CO2: Explain the current state and trends of technological growth, its contribution to human development and its use as an instrument of change in the 21st century;

CO3: Assess the impact of technology on environment and the impact of the environment upon human changes in the global climates;

CO4: Identify the environment friendly technologies, inc. renewable energy and environments and explain how they are environment friendly.

CO5: Identify the major hazards of industry and illustrate the ways to improve the working conditions in the industry and society.

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Technology, Environment & Society

Course Contents

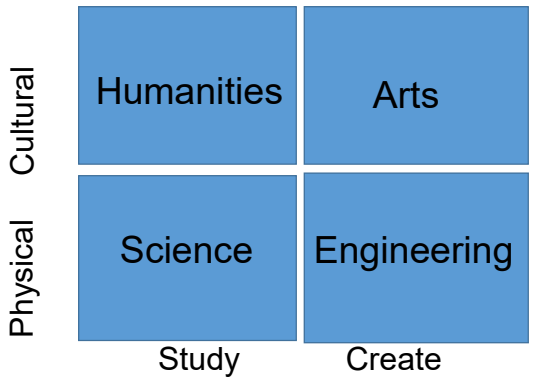
- Definition of terminology – technology, environment, society and development; Inter-dependence of technology, environment, society and development;
- Growth of technologies and its contribution to human development; Current state of technology and its future use as an instrument of change in twenty first century;
- Impact of technology upon the environment, impact of the environment upon human changes in the global climates;
- Environment friendly technology, Technology and development; Renewable energy and environments;
- Technology and environment hazards, its remedy. Major hazards of industry. The improvement of working conditions in the industry.

3

Technology, Environment & Society

Engineering and Technical education

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Engineering is
the discipline focused on
creating our physical world.

Projection of different disciplines onto a plane defined by 'cultural' vs. 'physical' and 'study' vs. 'create'.

(Src. Kapranos 2012).

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Technology, Environment & Society

Introduction

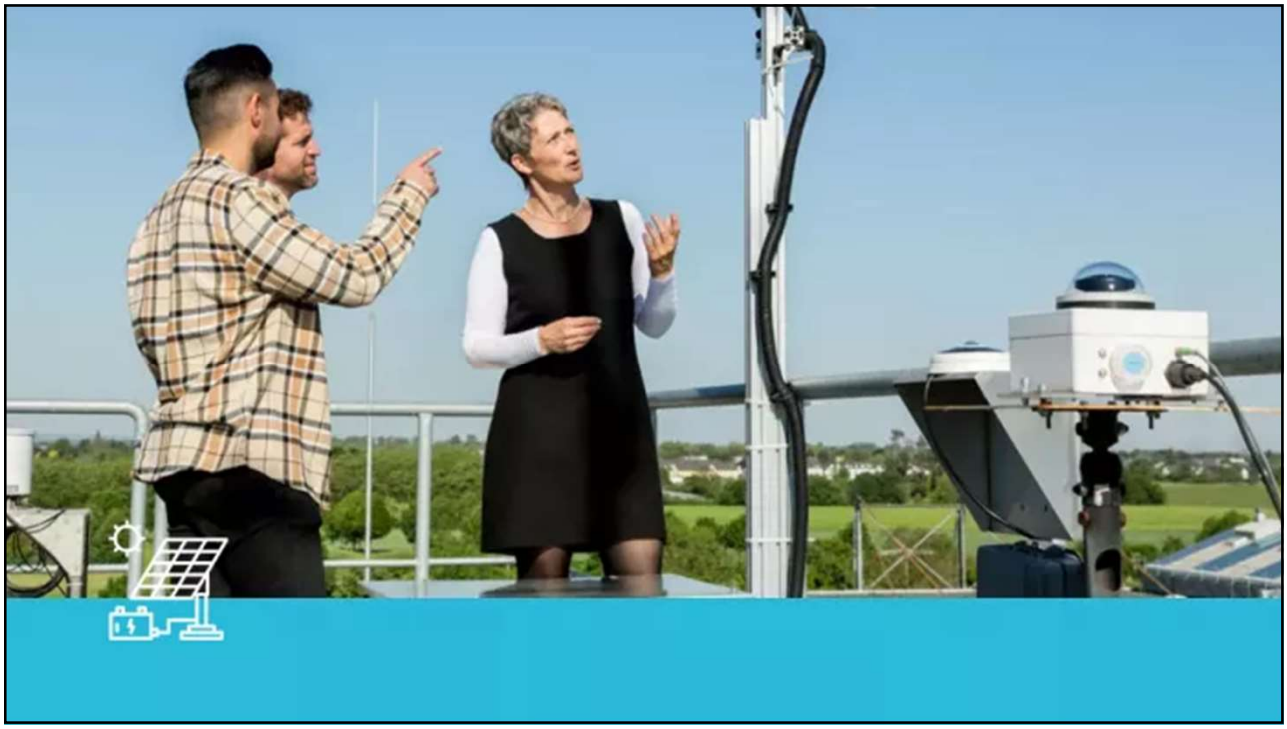
‘What do we envisage the engineers/technologists of the 21st century ought to be like’?

‘What competences (knowledge, skills, attitudes) should they possess’?

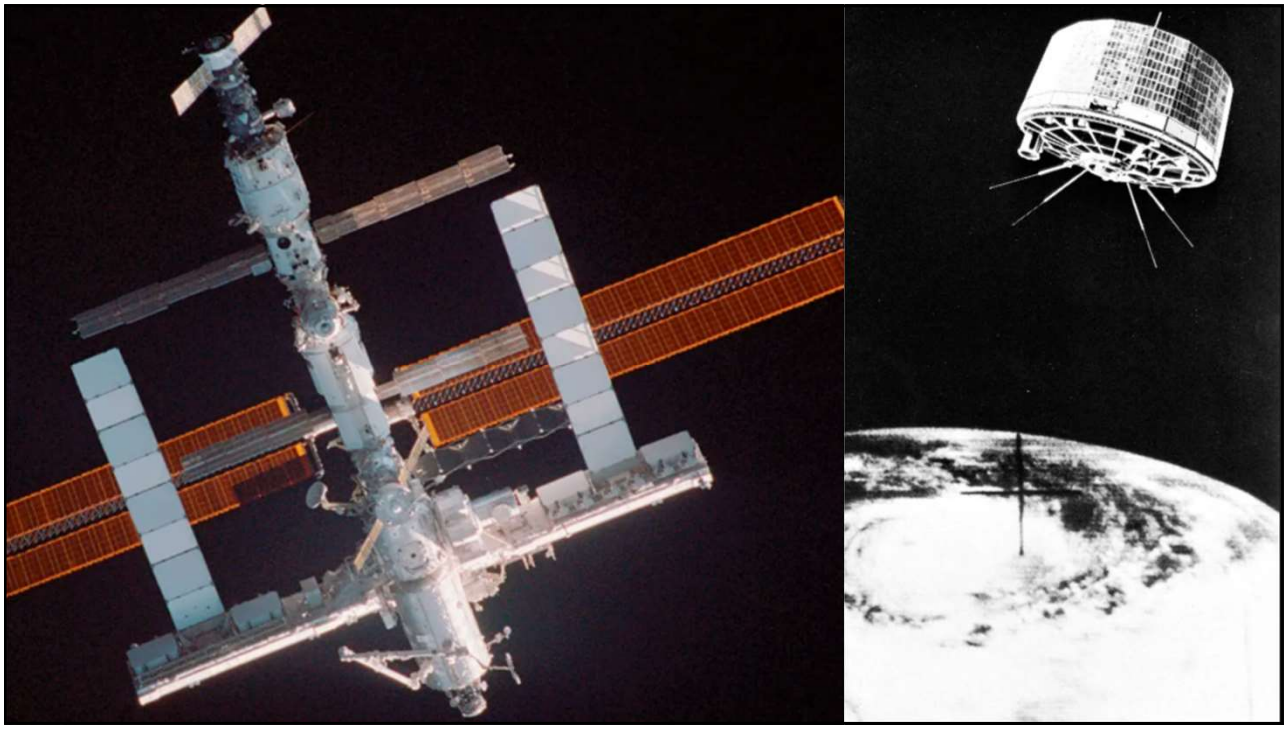




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After completing this session, you will be able to explain the interdependency among

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Technology, environment, society and development

- ✓ **(the study and knowledge of) the practical, especially industrial, use of scientific discoveries:**
  i.e. computer/medical/space technology.
- ✓ **the methods for using scientific discoveries for practical purposes,**
- ✓ **esp. in industry**
- ✓ **the use of scientific knowledge or processes in business,**
- ✓ **industry, manufacturing, etc.:**

https://dictionary.cambridge.org/dictionary/english/technology
https://en.wikipedia.org/wiki/Technology_and_society

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Technology, Environment & Society

What is technology?

The term ***technology***, a combination of the Greek ***technē***, "art, craft," with ***logos***, "word, speech," meant in Greece a discourse on the arts, both fine and applied.

Technology is

the application of scientific knowledge

to the practical aims of human life or,

as it is sometimes phrased,

to the change and manipulation of the human environment.

Is Internet technology "making us stupid"?

What is the impact of artificial intelligence (AI) technology on society?

https://www.britannica.com/technology/history-of-technology

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Technology, **Environment**, society and development

Definitions from Oxford Languages

✓ **the surroundings or conditions** in which a person, animal, or plant lives or operates.

Ex. "survival in an often hostile environment"

✓ **the natural world**, as a whole or in a particular geographical area, especially as affected by **human activity**.

Ex. "the impact of pesticides on the environment"

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Hum 4147: Technology, Environment and Society

Technology, **Environment**, society and development

Environment (NATURE)

✓ **the air, water, and land in or on which people, animals, and plants live:**

i.e. We're not doing enough to protect the **environment** *from pollution.*

Environment (SURROUNDINGS)

✓ **the conditions that you live or work in and the way that they influence how you feel or how effectively you can work:**

Studies show that if a working environment is pleasant, productivity increases.

https://dictionary.cambridge.org/dictionary/english/environment
https://en.wikipedia.org/wiki/Technology_and_society

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Technology, Environment & Society

Video1 Video1



Q: Watch these Videos and describe how technology and environment are interdependent to each other. Does technology change the ways of living/ lifestyle? Explain How?

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Technology, **Environment**, society and development

From Wikipedia, the free encyclopedia

Environment most often refers to:

Natural environment/Biophysical environment, referring respectively to all living and non-living things occurring naturally and the physical and biological factors along with their chemical interactions that affect an organism or a group of organisms

https://en.wikipedia.org/wiki/Environment

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Technology, Environment, **Society** and development

Society (PEOPLE)

- ✓ **a large group of people who live together in an organized way, making decisions about how to do things and sharing the work that needs to be done. All the people in a country, or in several similar countries, can be referred to as a society**

i.e.We must also consider the needs of the younger/older ***members of society****.*

- ✓ **people considered as a group, or a group of people who live together in a particular social system:**

i.e. Societies change over the course of time.

https://dictionary.cambridge.org/dictionary/english/society
https://en.wikipedia.org/wiki/Technology_and_society

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Technology, Environment, **Society** and **Development**

Assignment 1:
Explain how technology, environment, society and development are inter-dependent to each other. You may also refer to the videos showed in slide 13.

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Technology, Environment, Society and Development

History of Technology Timeline

3.3 million years ago: The first tools

6000 BCE: Irrigation

4000 BCE: Sailing

850 CE: Gunpowder

1 million years ago: Fire

1200 BCE: Iron

950: Windmill

1044: Compass

1250–1300: Mechanical clock

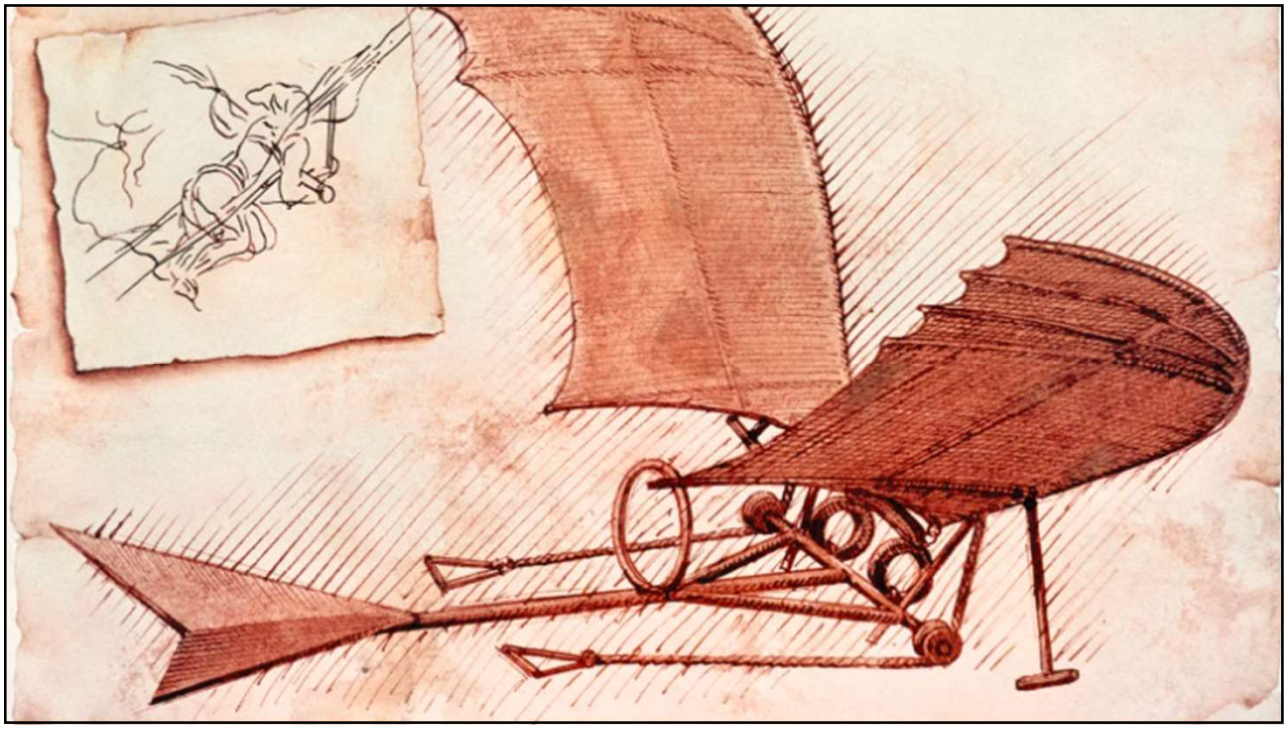
1455: Printing

1765: Steam engine

1804: Railways

https://www.britannica.com/story/history-of-technology-timeline

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Technology, Environment, **Society** and development

Relationship among Environment, Society and Technology

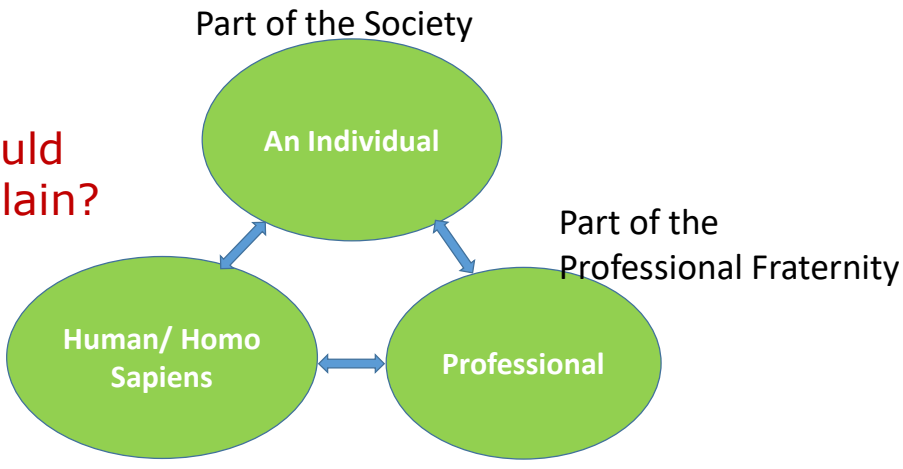
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Relationship among Technology, Environment, and Society

How would you explain?



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Relationship among Technology, Environment, and Society

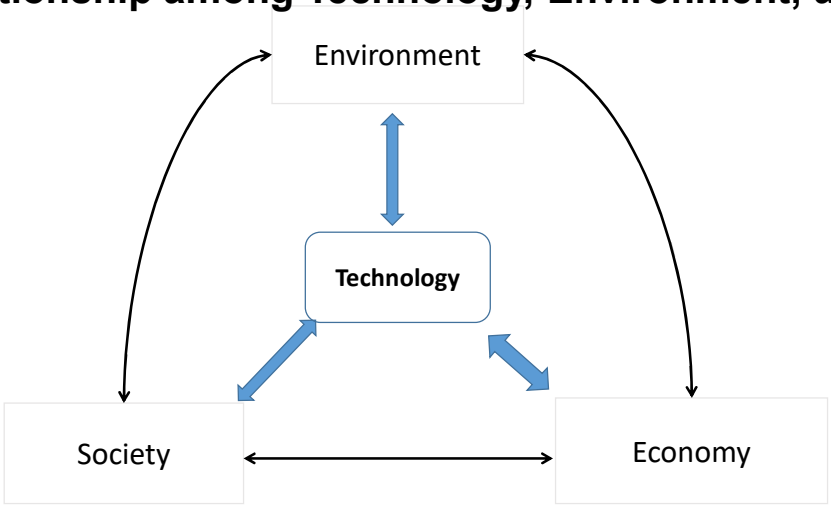
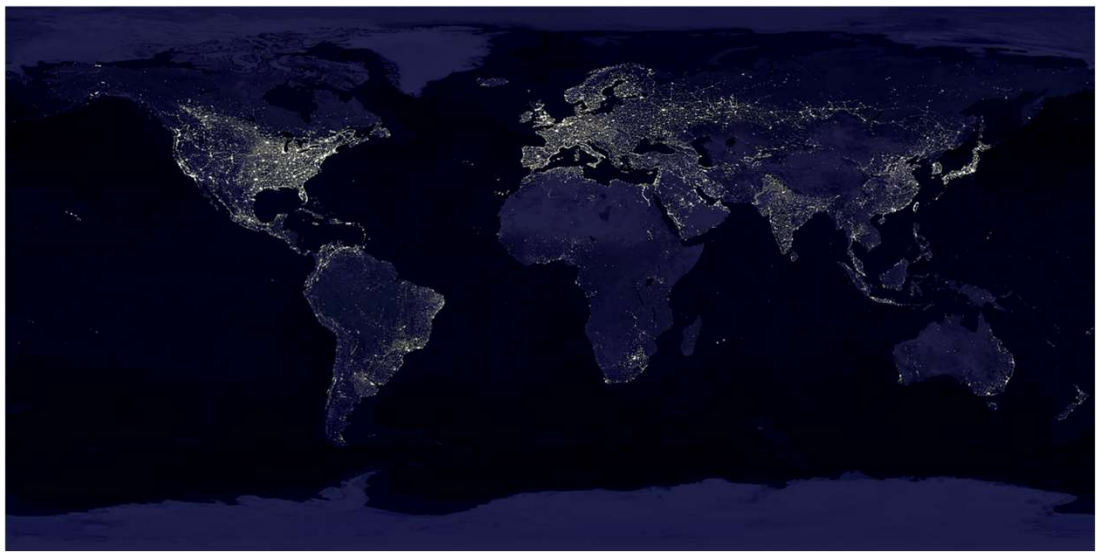


Fig.: Dimension of Sustainable Triangle

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NASA Image, 2012

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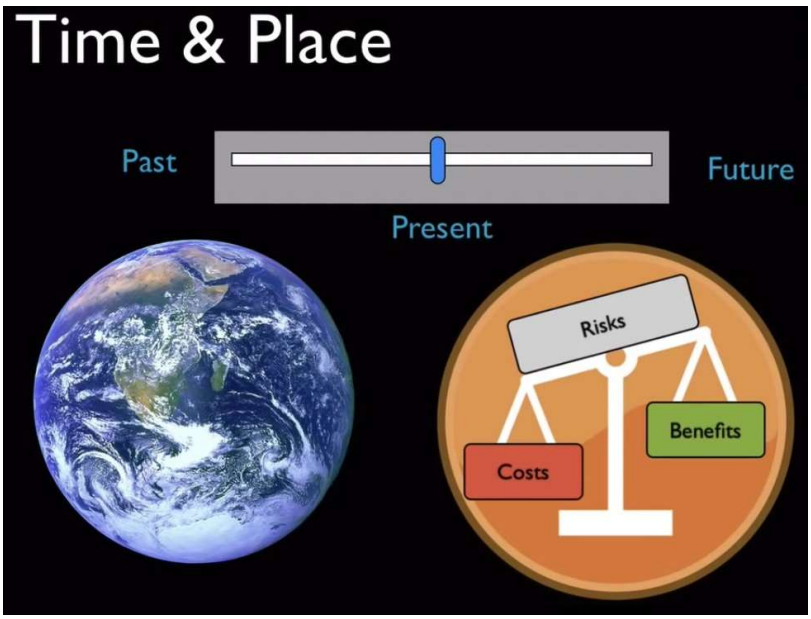
Relationship among Environment, Society and Technology



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Environment and the Earth System

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Environment and Earth System

Environment

The **environment** is the aggregate of all external factors or conditions that influence the activities and existence of all living beings including human. Or

The environment is the interaction within and among physical, living and human world in our **surrounding.**

Surroundings!!!

- A certain distance?
- the room; we live in?
- building?
- City?
- District or division?
- Country?
- Planet?

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

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Environment and Earth System

The environment has three important constituents or worlds:

1. Physical world/ Abiotic; consists of nonliving objects (e.g. Rocks, soil, cloud, air) and processes (e.g. Evaporation, season, weather, climate etc.).

This **physical constituent** of the environment is again divided into three parts: (i) **Atmosphere** (gas), (ii) **Hydrosphere** (liquid), and (iii) **Lithosphere** (solid)

2. Living world/ Biotic; consists of living things (e.g. millions of plants, animals, fungus and bacteria's) and life process (e.g. feeding, reproduction species, respiration and photo-synthesis, etc.)

3. Human world/Social; consists of all things that human make, do and believe (e.g. science, arts, religion, literature etc.)

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Environment and the Earth System

Q: Do these three 'worlds' exist in total isolation from one another?

Ans: Of course not!!!

Interaction between PW, LW and HW

Inhale ⟶ Air (PW)

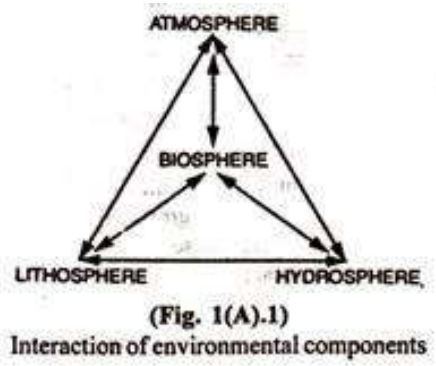
Eat ⟶ Food (LW)

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

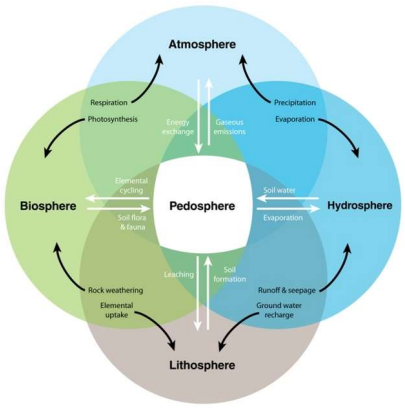
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Environment and the Earth System

The three components of the environment give rise to four important zones: 1. Atmosphere, 2. Hydrosphere, 3. Lithosphere and 4. Biosphere.



(Fig. 1(A).1)
Interaction of environmental components



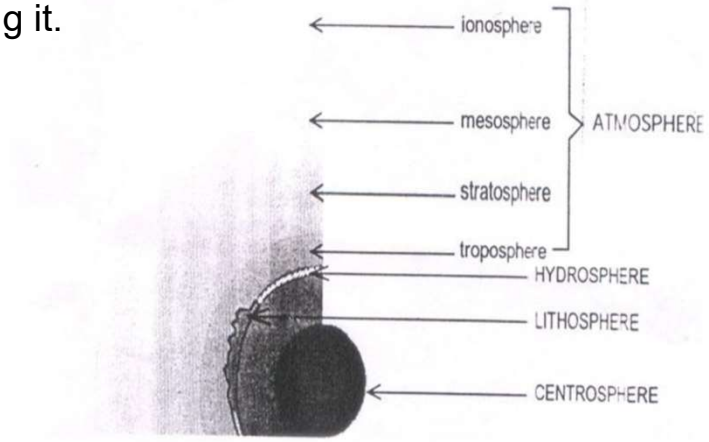
There is continuous interaction among these four zones.

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Environment and Earth System

Earth System

The earth is a complex system consisting of different components; the lithosphere, the hydrosphere, the atmosphere and the biota inhabiting it.



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Environment and **Earth System**

Centrosphere

Earth was once a **molten mass** and to this date, the temperature and pressure in the interior of the earth is so high that the molten state still exists there. These are brought to the surface from time to time as **lava**, through **volcanic eruption** or through faults in the earth's crust.

Lithosphere

The lithosphere consists of the **solid crust** on which rest the continental masses comprising of **mountains, valleys and plains**. This is where human beings and other terrestrial animals live. The soil covering the solid masses in most cases is fertile enough to support plant life and in its natural state is covered with vegetation.

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Environment and **Earth System**

Hydrosphere

The hydrosphere consists of these oceans, seas and the inland water bodies. **Ninety-seven percent** of the total supply of water on earth are held in the oceans and seas whilst the rest is freshwater. Life on earth would not be possible without water. All the life supporting processes of the earth system are heavily dependent on water.

Atmosphere

The atmosphere is a gaseous mantle enveloping the earth. The main constituents are nitrogen (78%) and Oxygen (21 %). The rest 1 percent is mostly argon, carbon dioxide and some other trace gases.
It consists of different strata called: Troposphere, Stratosphere, Mesosphere and Thermosphere.

Biota: all living organism including Flora and Fauna

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Ecology and **Ecosystem**

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Ecology and Ecosystem

is the study of **living organisms** in the **natural environment**.
How they interact with **one another** and
how they interact with their **non living environment**.

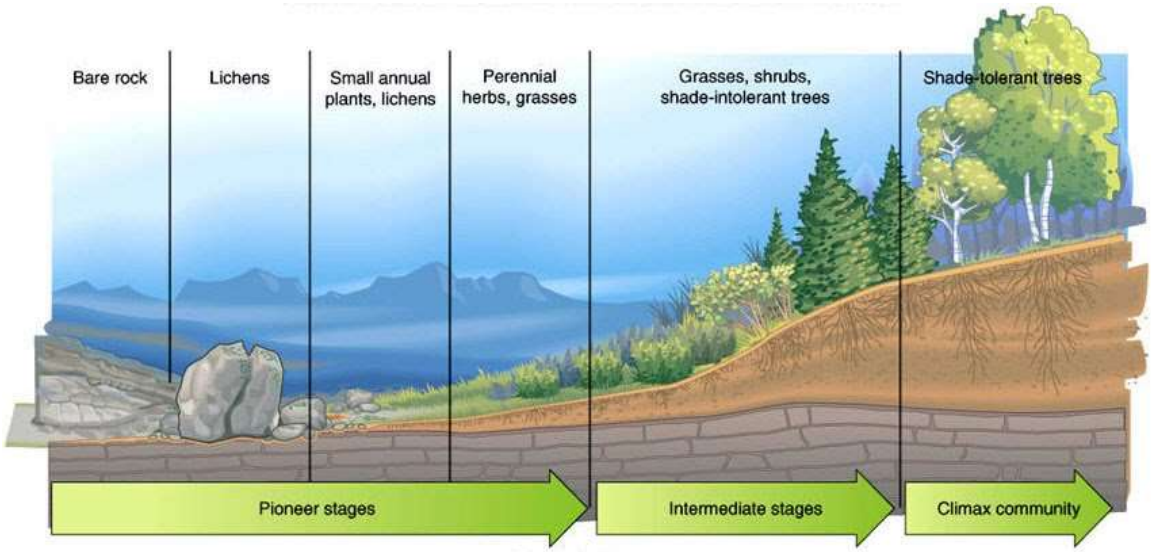
is the relationship between organism and environment. (Hackel and Tansley)

Terminologies
Autecology: Individual (Study of RB Tiger Population)
Synecology: Group/community (Study of Forest Ecosystem)
Ecological Succession: **process of change** in the species structure over time
Ecological Niche: Physical + Functional **role of a species** in a community
Ecological Equivalent: Similar niche in different geographical regions

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Ecological Succession



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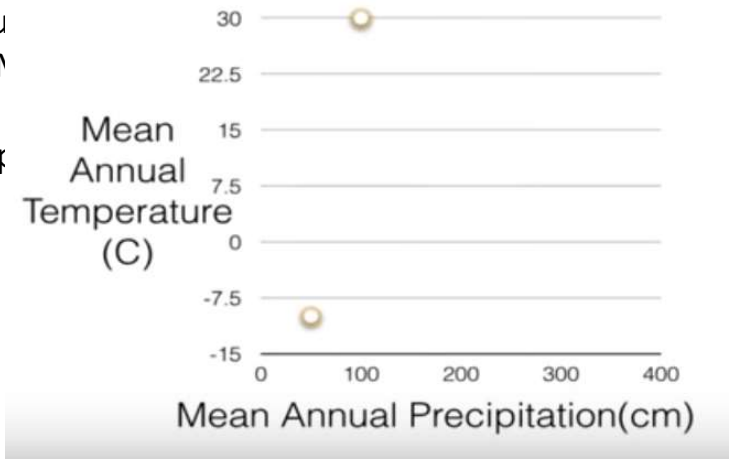
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Ecological Succession

Therms:

- Mega: High, T. Rain Forest
- Meso: High+Low, T. Decidu
- Micro: Low Temperature, M Coniferous
- Hekisto: Extremely low, Alp

Biomes:
very large ecological areas; climate, vegetation, soil



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Ecological Succession

A biome is determined mainly by its climate-like temperature and rainfall.
The earth's land areas are divided into **6 major biomes**.

1. TUNDRA

- Bitterly cold and covered with snow and ice.
- Ground remains permanently frozen (permafrost).
- Only certain small plants such as mosses and grasses can grow.



2. CONIFEROUS FOREST (TAIGA)

- Confers = cone-bearing trees (pines, fir)
- Cold climate
- Dense forests (trees tops block out a lot of sunlight)
- Grasses, small trees cannot grow – only some shrubs, ferns and mosses thrive.
- Animals such as squirrels, moose (sambur), birds, and insects live here.



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Ecological Succession

3. DECIDUOUS (TEMPERATE) FOREST

- Thrive in moderate climates.
- Summers may be hot and winters may be cold.
- Dense forests
- Provides habitats for many animals.



4. TROPICAL RAIN FOREST

- Excellent for plant growth and soil development.
- Plants grow thick and tall.
- Found in areas near the equator.
- Support more plant life and animal species than any other biome.



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Ecological Succession

5. GRASSLANDS

- Mostly grass!
- Temperatures similar to deciduous forest.
- Do not receive much rainfall –enough rain to support grasses but not trees.



6. DESERT

- Very dry; receives very little rainfall.
- Very hot during the day, cold at night.
- Soil is very dry and poor.
- Only a few kinds of plants grow in the desert.
- Very few animals can survive in the desert.



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Ecological Succession

Species

A group of organisms that can breed to produce fully fertile offspring



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Ecological Succession

Population

A group of organism of the same species which live in the same habitat at the same time where they can freely interbreed.



The black-veined white butterfly *(Aporia crataegi)* mating

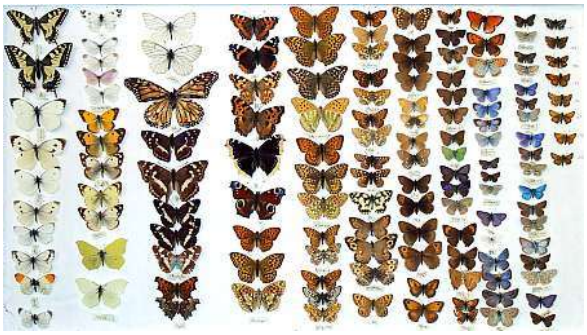
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Biodiversity

The total number of different species in an ecosystem and their relative abundance





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The characteristics of the type of environment where an organism normally lives. (e.g. a stoney stream, a deciduous temperate woodland, Bavarian beer mats)

Habitat



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Ecology and **Ecosystem**

A unit which consists of

all living organisms functioning together in a given area and interacting with the physical environment in a manner, whereby, flow of energy leads to clearly defined biotic structures and cycling of materials between living and non-living parts, is know as an ecosystem.

- Ecosystem, the term coined by Tansley
- Smallest structural and functional unit of nature
- **Consisting of both biotic community and the abiotic environment which have close interactions**
- The interaction is conducted by **energy flow** (**Solar energy**) in the system and cycling of materials (**Nutrient cycle**)
- Entire biosphere is considered as a global ecosystem

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Ecology and **Ecosystem**

https://www.britannica.com/science/ecosystem

An ecosystem can be categorized into its abiotic constituents, incl. minerals, climate, soil, water, sunlight, and all other **nonliving elements**, and its **biotic constituents**, consisting of all its living members.

Two major forces link these constituents together:
(i) the flow of energy through the ecosystem and
(ii) the cycling of nutrients within the ecosystem.



Community + Abiotic environment, interacting

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Ecology and **Ecosystem**

Characteristics of Ecosystem

Boundaries are indistinct and over-lapping; could be temporary or permanent; even may be small (drop of water) or large (whole earth)

Types of Ecosystem: on the basis of the origin

1.**Natural** Ecosystem; terrestrial (forest, grassland and desert) and aquatic (pond, lake, wetland, river and estuary).

2. **Man-made** Ecosystem; crop field and aquarium.

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Ecology and **Ecosystem**

Types of Ecosystem: **on the basis of the size**

- ❑ Mega Ecosystem: Entire Biosphere
- ❑ Macro Ecosystem: Major Biomes like Tropical Rain forest (Amazon)
- ❑ Micro Ecosystem: Pond
- ❑ Nano Ecosystem: Drop of water

COMPONENTS OF ECOSYSTEM

1. **Biotic Components** includes living component from smallest microbe, fungi to largest animals and plants
2. **Abiotic Component** includes Inorganic substance (carbon, nitrogen, CO_2, H_2O etc.) involved in natural cycles, air, water and other physical factors, organic compounds (Proteins, Carbohydrates, Humus)

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Ecology and **Ecosystem**

Among all abiotic components; the important ones are:

- **Temperature**; spatial distribution of species decreases on either side of equator towards the poles and as we go above and below the MSL.
- Water; Life started in water about 3.5 billion years ago.
- Light; photosynthesis
- Soil; Edaphic factors. Based on formation process, it is classified in two types:

  Residual soil – Weathering and pedogenesis occur at the same place
  Transported soil - Weathering and pedogenesis occur at different places (Colluvial- Brought by gravity, Alluvial- brought by water, Glacial- by snow, Eolian- by wind)

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Structure and Function of Ecosystem

Vertical distribution species occupying different levels is called stratification. For instance, trees canopy is top layer of a forest while grasses are at the bottom.

Functions:

a) Productivity b) Decomposition c) Energy Flow and d) Nutrient Cycle

a) Productivity

Biomass: Mass of living biological organism in a given area at a given time

species biomass-mass of one species

community biomass- mass of all species in the community (including microorganisms, plants or animals). It is measured either in terms of dry weight (60-80% weight of water not included) or by measuring organically bound carbon.

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Structure and Function of Ecosystem

Primary Production: Total amount of biomass or organic matter produced per unit area. Unit in terms of weight (gm^{-2}) or energy ($kcal/m^2$)

Primary productivity: Amount of biomass produced per unit area over a time period by plants during photosynthesis ($gm^{-2}y^{r-1}$/$kcalm^{-2}yr^{-1}$)
Secondary Productivity is defined as the rate of formation of new organic matter by consumer

b) Decomposition

Decomposers break down complex organic matter into inorganic substances like CO_2, water and nutrients and the process is called decomposition.

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Structure and Function of Ecosystem

Primary Production: Total amount of biomass or organic matter produced per unit area. Unit in terms of weight (gm^{-2}) or energy ($kcal/m^2$)

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Decomposition
Decomposers break down complex organic matter into inorganic substances like CO_2, water and nutrients and the process is called decomposition.

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Structure and Function of Ecosystem

Energy is transferred from one trophic level to another

- First trophic level (T1) = Producer (Phytoplankton, Grass, Trees)
- Second trophic level (T2) = Primary consumer (Zooplankton, rabbit)
- Third trophic level (T3) = secondary consumer (Birds, fishes)
- Fourth trophic level (T4) = top consumer (Man, Lion)

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Structure and Function of Ecosystem

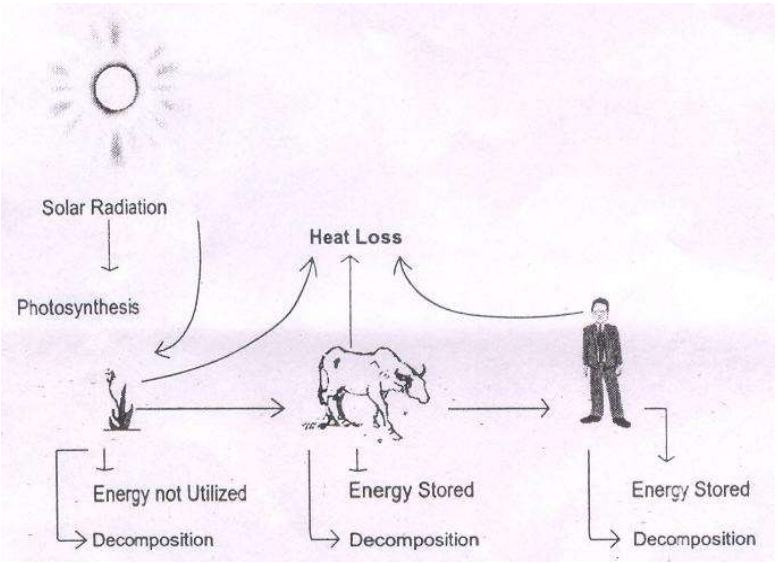
Rule of 10% (10% law of Lindeman): energy is reduced by 10% during flow to every higher trophic level. Because of this law, the number of trophic levels are usually restricted to 4-5.

Right from the sunlight being trapped by autotrophs into organic material, and further transfer of energy to higher trophic levels. On an avegrage, 10% (2-20%) of energy is transferred while rest 90% (80-98%) is lost in respiratory and other losses. So as the number of trophic levels increase in a food chain, most of the energy is lost.

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Energy Flow in the Food Chain

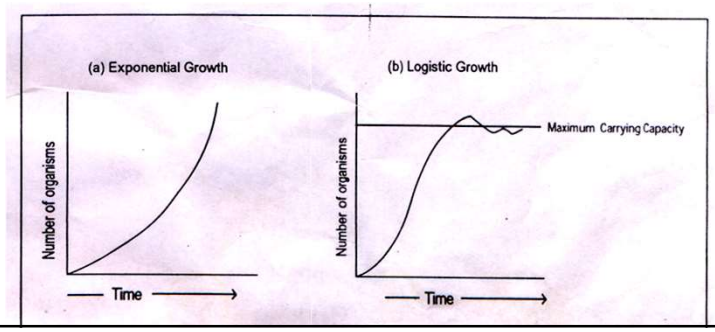


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Carrying Capacity

Carrying capacity of environment or an ecosystem is the **threshold limit** of use of that system without damaging it.
Every ecosystem has its resources that are used for economic development, for survival and for habitat creation. Environment and ecosystems have got the abilities to recover the loss of its resources by regenerating them over the period of time that are temporary and not exceeding the threshold damage limit.



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Carrying Capacity

Catton (1986), defined an environment's carrying capacity as its maximum persistently supportable load.

Ecological carrying capacity is a measure of the amount of renewable resources in the environment in units of the number of organisms these resources can support. Normally, K is a function of both the species and the environment, and is expected to change only in evolutionary time (Daily and Ehrlich, 1992).



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Carrying Capacity



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Sustainable development (see other slides);

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Thank You!
Welcoming Q & A

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